also train adversarial perturbations against only the model output. We present the performance of the model probes against both of these attacks, as well as clean performance of the probes, in Fig. 16.

Image classification latent probes are not robust to attacks. Our results concerning standard attacks match those of Fort and Lakshminarayanan [2024]: early layer probes maintain performance under attacks against the model output, with performance decreasing in later layers. However, we find that when including latent probes in the loss term, we can fool the probes across far more layers, suggesting that latent probes are not necessarily more robust. These findings are in line with concurrent work Schlarmann et al. [2024], Zhang et al. [2024a], which like us attacks the mean prediction of all probes and shows that this can fool the entire ensemble.

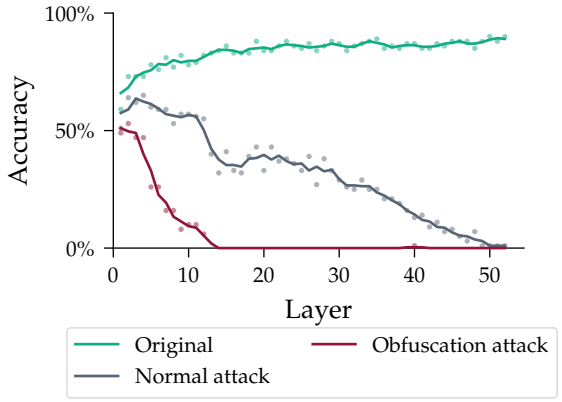

Figure 16: **Image classifiers are vulnerable to obfuscated representations.** Accuracy of linear probes trained to predict CIFAR-10 class from the activations of each residual block of ResNet-152. Performance on **clean images** increases as a function of layers. **Output-only attacks** decrease accuracy in later layers, but only partially in early and middle layers. **Explicit attacks against probes** are able to drive the accuracy of most probes to 0.

6.2 Training models to be easier to monitor

Throughout this work, we have seen many latent monitoring methods fail against obfuscation attacks. Even adversarial training did not solve this issue (§3.5). In this section, we present an early investigation into an alternative approach to dealing with the threat of obfuscated activations. Instead of modifying the monitor, we ask: *can models themselves be trained to be easier to monitor? And how does this affect robustness against obfuscation attacks?*

Inspired by Sheshadri et al. [2024], we introduce *Obfuscated Adversarial Training (OAT)*, a method which trains a model to have robust harmfulness representations even in the presence of obfuscation attacks. OAT first trains a monitor D as in earlier sections. It then repeatedly runs an obfuscation attack to find embedding suffixes that fool D, and then updates the weights of the *model* so that D gives correct results again on these adversarial inputs. The monitor D stays frozen during that process.

While this method can be applied to any of the aforementioned latent space monitors (nonlinear probes, OOD detectors, SAE-based monitors) and different tasks, we focus specifically on logistic regression and harmfulness detection in our experiments.

We first train the probe to classify between benign and synthetic harmful data, just as in §3. We then apply OAT (Algorithm 1), i.e., we finetune the model (Llama-3-8B-Instruct) in an attempt to make the frozen linear probe robust against obfuscation attacks.

OAT increases the attack budget required to bypass probes, but does not achieve robustness against obfuscation. Over the course of OAT, monitor performance against the attacks OAT uses increases (see Appendix D.1). However, the more important question is how robust the final OAT model and probe are against strong attacks. We find that an obfuscation attack with

a universal embedding suffix (as in §3) achieves a jailbreaking rate of 35% and zero recall (at 1% FPR) after 16,384 gradient steps (about 10 hours on a single GPU).[6] In comparison, against Llama-3-8B-Instruct without OAT, the same obfuscation attack achieves a 95% jailbreak rate and zero recall after only 256 gradient steps (or about 10 minutes). Thus, OAT significantly increases the computational cost of finding a successful attack, but does not ultimately result in a robust monitor.

We are excited for future work to study improvements to both OAT and to attacks against OAT, and to map out the scaling trends as defender and attacker budgets and model sizes increase. Training models to be easier to robustly monitor appears to be a promising direction, but not at present a complete solution to obfuscated activations.

6.3 Limits of obfuscation

In this section, we explore just how malleable representations are. Considering a model processing inputs from UltraChat [Ding et al., 2023], we ask: *can a universal embedding suffix preserve the model's behavior while making its activations orthogonal to unperturbed ones?* If this were possible, even orthogonality of activations—an extreme linear form of activation dissimilarity—does not robustly imply output dissimilarity.

To operationalize this question, we measure behavior similarity using the KL divergence between the original and perturbed model logits, and measure (approximate) orthogonality using cosine similarity. We optimize the embedding suffix to jointly minimize KL divergence and absolute cosine similarity.

Figure 17 shows the trade-off between preserving the behavior (low KL divergence) and obfuscation (low cosine similarity) as we vary the obfuscation loss weight, the length of the suffix, and the learning rate.

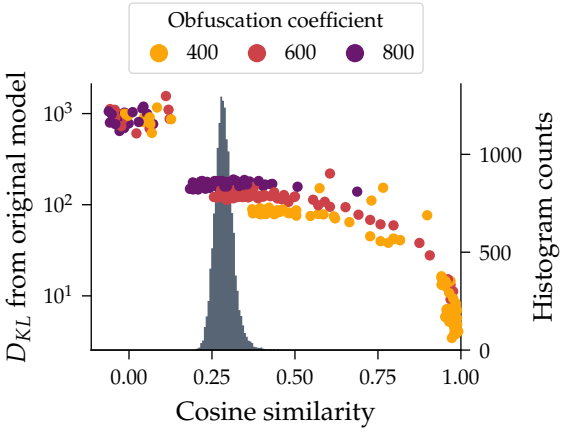

Figure 17: **There is a trade-off between making representations orthogonal (low cosine similarity) and retaining behavior (low D_{KL}).** Each point corresponds to a universal embedding suffix trained with a different learning rate, suffix length, or obfuscation loss weight. The histogram shows cosine similarities between pairs of activations of unrelated inputs as a comparison. a D_{KL} of 100 qualitatively leads to strong behavior deterioration, so *we don't achieve cosine similarity comparable to unrelated inputs while maintaining behavior.*

We are unable to achieve near-orthogonality while preserving behavior. Activations of two randomly chosen unrelated inputs from UltraChat have a cosine similarity of about 0.3, and to reach that level of orthogonality, we need to accept a KL divergence of over 100. Qualitative inspection of those samples shows that this KL divergence corresponds to mostly incoherent output.

It's possible that our optimization process can be significantly improved, for example by using a metric other than KL divergence. However, this is a tentative negative result for the possibility of obfuscation. It shows that even

[6] We expect that a higher jailbreaking rate is possible by weighting the jailbreaking loss more strongly relative to the obfuscation loss, or training the embedding even longer.

embedding attacks can not achieve *arbitrary* results in the latent space if we simultaneously want to achieve certain behaviors.

7 Related Work

Mechanistic Defenses LLMs have tended to be persistently vulnerable to a wide variety of adversarial attack techniques that jailbreak the model or otherwise elicit harmful behaviors [Carlini et al., 2024, Shayegani et al., 2023, Andriushchenko et al., 2024, Łucki et al., 2024, Geiping et al., 2024]. Adversarial fine-tuning techniques have shown a very limited ability to make models robust to existing attacks [Anwar et al., 2024]. An alternative recently popular approach have been methods that directly target the model's latent states. This has included:

- *Representation engineering* approaches, such as classifiers probing for concepts (e.g. harmfulness) from model activations [Alain and Bengio, 2016, Belinkov, 2022, Zou et al., 2023a, Mallen et al., 2023, Arditi et al., 2024, Ball et al., 2024, Bricken et al., 2024, MacDiarmid et al., 2024], or techniques for finding steering vectors [Todd et al., 2023, Turner et al., 2023, Panickssery et al., 2023, Zhang et al., 2024b, Mack and Turner, 2024].
- *Unsupervised dictionary learning* methods for dividing latent space into interpretable directions, such as with sparse autoencoders [Yun et al., 2021, Cunningham et al., 2023, Bricken et al., 2023, Braun et al., 2024a, Gao et al., 2024, Templeton et al., 2024, Lieberum et al., 2024] or codebook features [Tamkin et al., 2023].
- *Latent adversarial training (LAT)* techniques for training models under adversarial perturbations to their hidden activations [Casper et al., 2024, Sheshadri et al., 2024].
- *Representation corruption* methods which train LLMs to corrupt their latent states when encountering harmful text [Zou et al., 2024, Rosati et al., 2024, Gandikota et al., 2024, Li et al., 2024].
- *Latent OOD detection* methods for flagging out-of-distribution representations [Podolskiy et al., 2021, Raghuram et al., 2021] such as those caused by adversarial attacks [Li and Li, 2016, Metzen et al., 2017, Feinman et al., 2017, Grosse et al., 2017, Amirian et al., 2018, Ma et al., 2018, Abusnaina et al., 2021, Zhao et al., 2023] or backdoored inputs [Chen et al., 2018, Tran et al., 2018, Gao et al., 2019, Tang et al., 2021, Hayase et al., 2021, Chen et al., 2022, Wang et al., 2022, Liu et al., 2022, Ma et al., 2023, Mo et al., 2024].

Evading Latent-Space Defense Methods While there has been significant work showing that adversarial attacks can avoid latent-space defenses in the image domain [Carlini and Wagner, 2017, Athalye et al., 2018, Hosseini et al., 2019, Kaya et al., 2022], adversarial attacks on LLMs [Yang et al., 2021, Wei et al., 2023, Carlini et al., 2024, Zou et al., 2023b] have so far mainly not investigated avoiding latent-space defenses, with a few recent exceptions. Although they did not optimize for obfuscation, Kirch et al. [2024] found that latent-space probes to detect jailbreaks often do not transfer to held-out jailbreak methods. Similarly, Arditi et al. [2024] and Jain et al. [2024] investigated the latent-space properties of jailbreaks. Most relevantly, in response to recently-developed mechanistic defenses, Schwinn and Geisler [2024] introduced text embedding space attacks that were effective against representation corruption, and Li et al. [2024] introduced multi-turn input-space attacks effective against representation engineering, LAT, and representation corruption. However, unlike either of these works, we study the problem of obfuscated activations